

Regulation of gene expression in eukaryotes

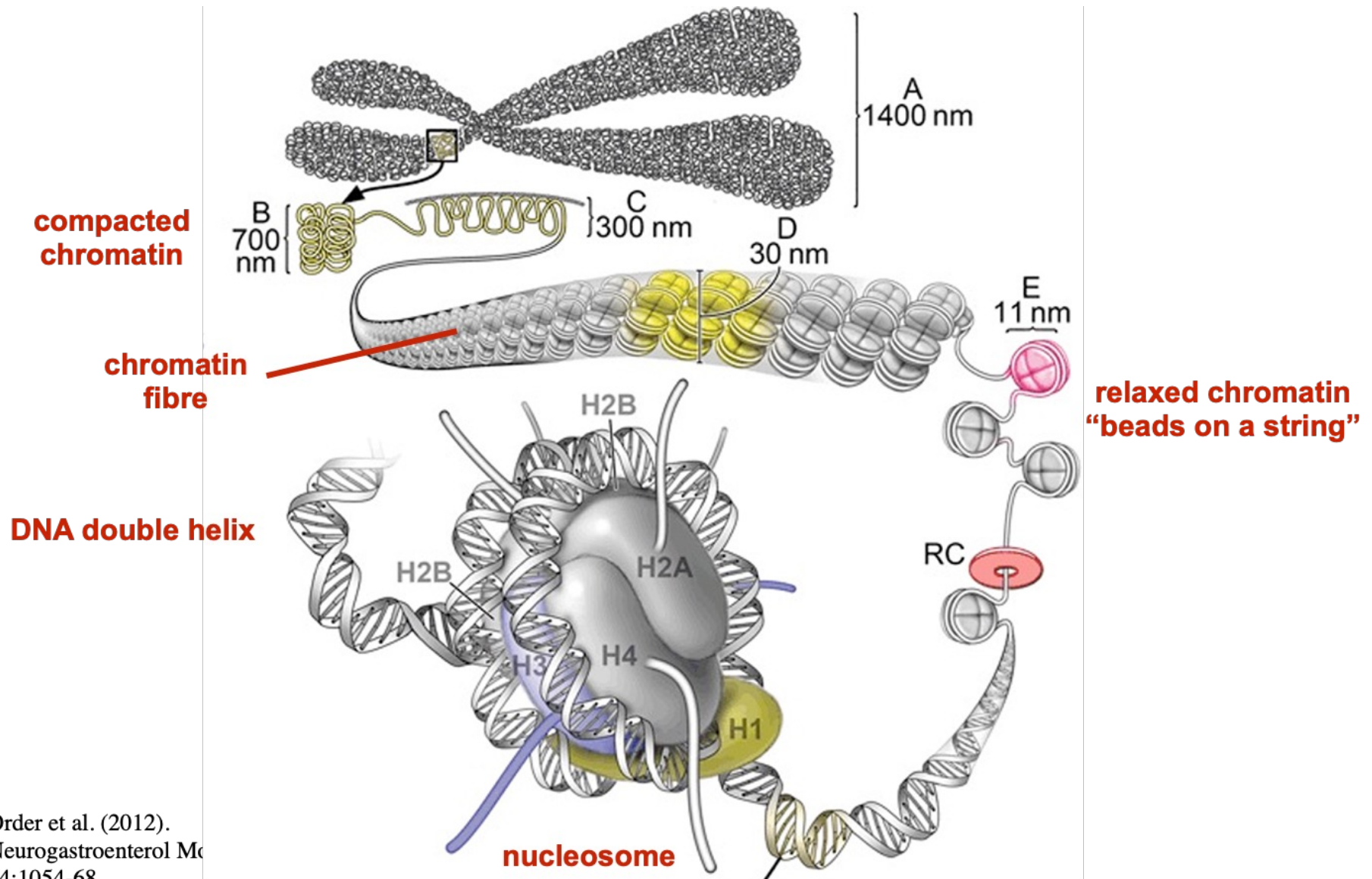
Readings:

Ch 8 277-282 [section 8.3]

Ch 12 399-405, 406-418 [sections Intro-12.3 (not yeast mating type)]

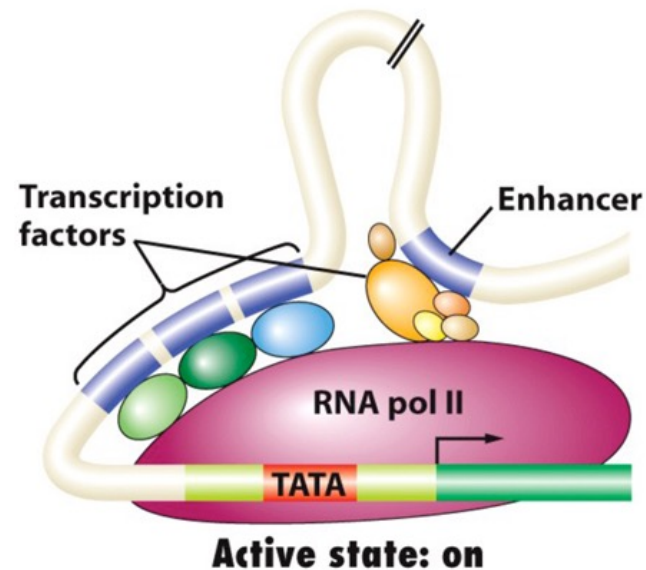
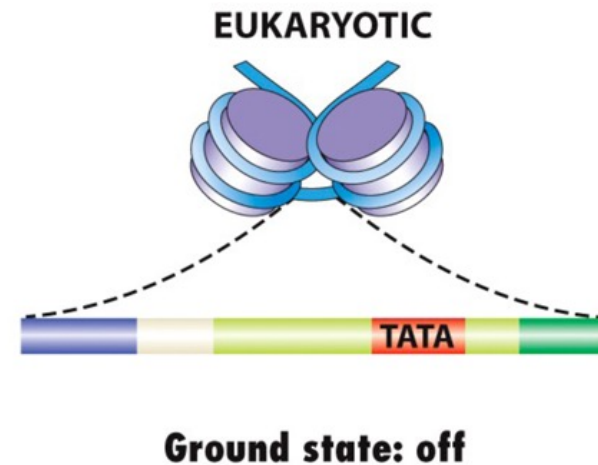
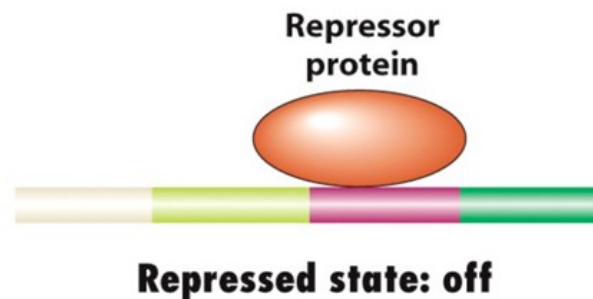
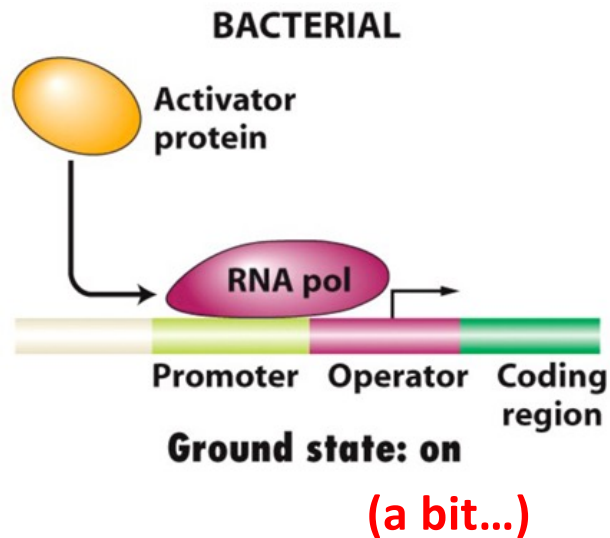
- Transcriptional control in eukaryotes
 - Promoter elements: core and proximal; enhancers/silencers
 - example – yeast *Gal* system
 - Transcriptional regulation by changes to chromatin structure
 - Position-effect variegation in *Drosophila*

DNA + associated proteins = chromatin



Order et al. (2012).
Neurogastroenterol Mo
24:1054-68

Difference between bacterial and eukaryotic transcription



Chromatin structure

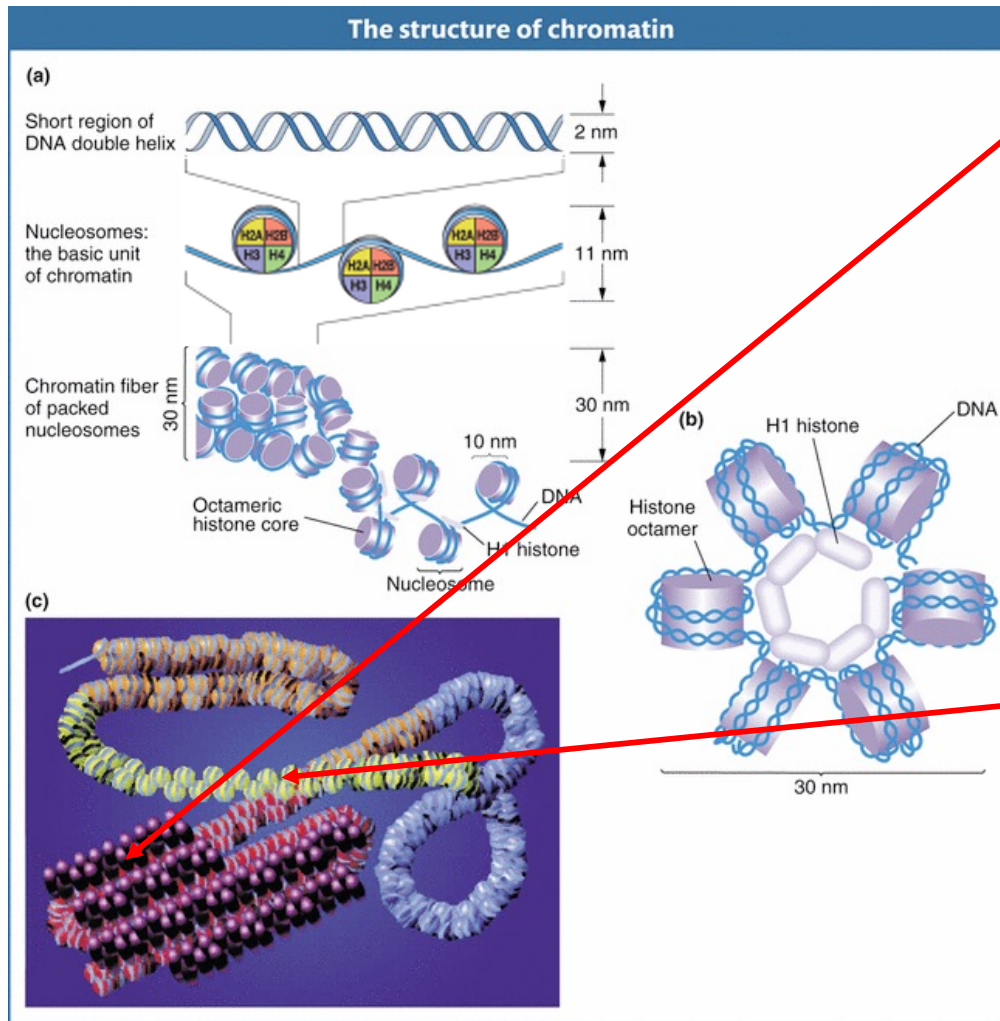


Fig 12-11

Heterochromatin

- tightly packed nucleosomes
- “closed” DNA
- associated with “structural” DNAs & non-transcribed regions
- e.g. centromere, telomeres, repetitive elements

Euchromatin

- loosely packed nucleosomes
- “open” DNA
- associated with transcribed regions of chromosomes

Any molecular processes that need to access chromosomal DNA (eg. transcription, replication, and repair) require that nucleosomes be “unpacked”.

Transcription initiation

- like in prokaryotes, it involves cis-acting DNA sequences and trans-acting proteins.

- ***cis-acting DNA sequences***

- core promoter
- promoter-proximal elements
- enhancers/silencers

- ***trans-acting proteins***

- general transcription factors
- common transcription factors
- cell/tissue-specific transcription factors
- transcription cofactors

Discussed in Primers and textbook readings

Transcriptional regulation of cell/ tissue-specific genes

Efficient transcription of such genes requires binding of specific transcription factors to **enhancer** sequences

- distance-independent *cis*-acting DNA sequences
- can be found near to or far from the transcription start site (TSS), situated upstream, downstream, or even within the gene itself (introns)
- **enhancers** – promote transcription
- **silencers** - prevent transcription

Transcriptional regulation of cell/ tissue-specific genes

Efficient transcription of such genes requires binding of **specific transcription factors** to enhancer sequences

- expressed in cell-type / tissue / time-specific manner
- bind DNA in a sequence-specific manner through DNA-binding domain
- influence transcription by:
 - interacting with transcription machinery (e.g. RNAPol) directly or indirectly OR
 - influence chromatin structure directly or indirectly

Eukaryotic transcription factors and cofactors

Transcription factors have multiple domains

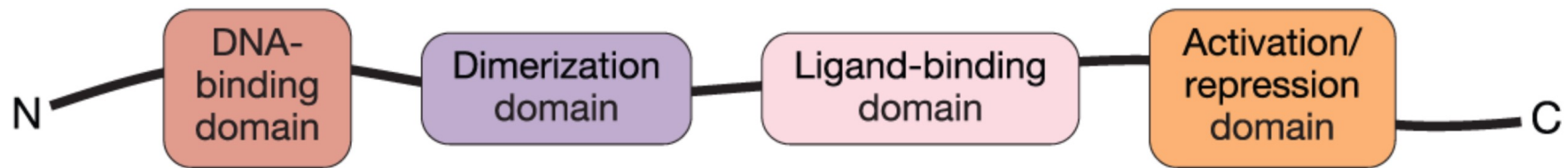
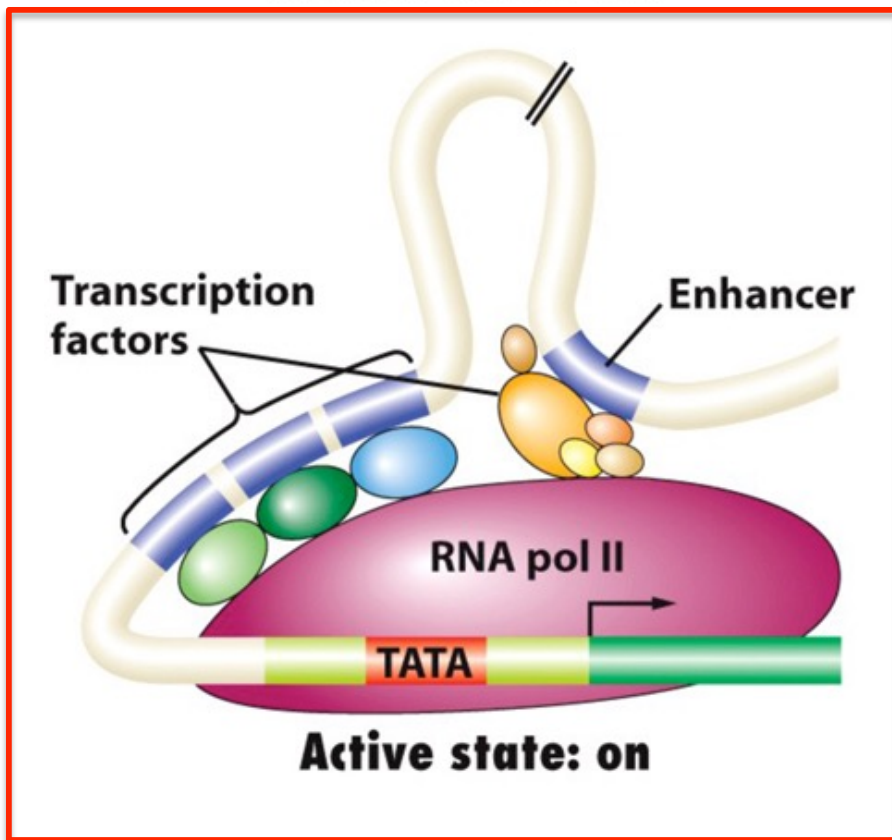


Fig. 12-6

- there are many different families of **transcription factors**; all have a sequence-specific DNA binding domain
- all have an activation or repression domain that mediates interactions with other components of the transcription machinery
- can have other domains – dimerization, ligand-binding domains (= sensors)
- transcription **co-factors** can have similar domains, but **LACK A DNA BINDING DOMAIN**
- function to regulate gene transcription

Enhancer-bound transcription factors help recruit RNAPol II to the promoter

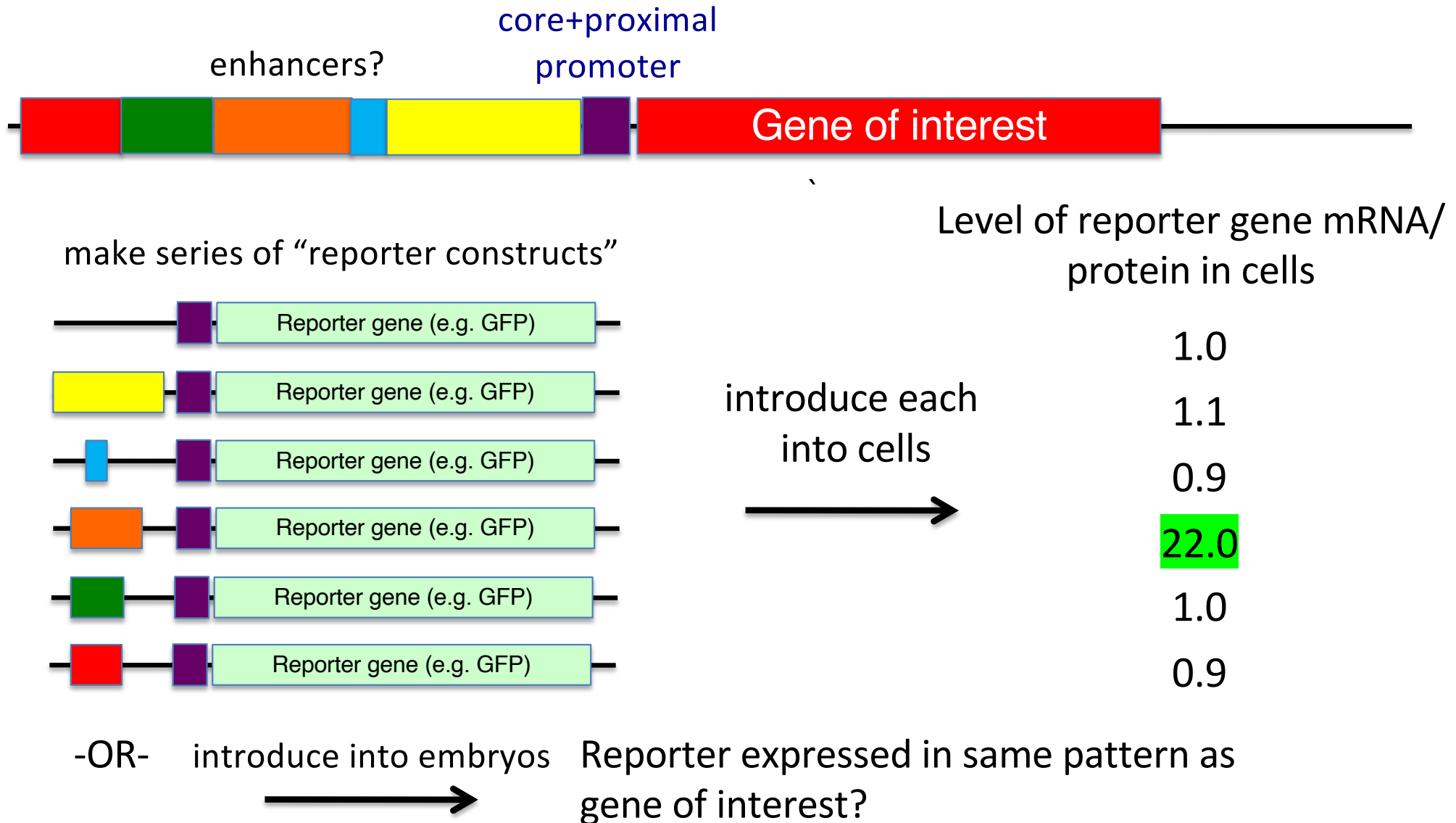
“ENHANCEOSOME”



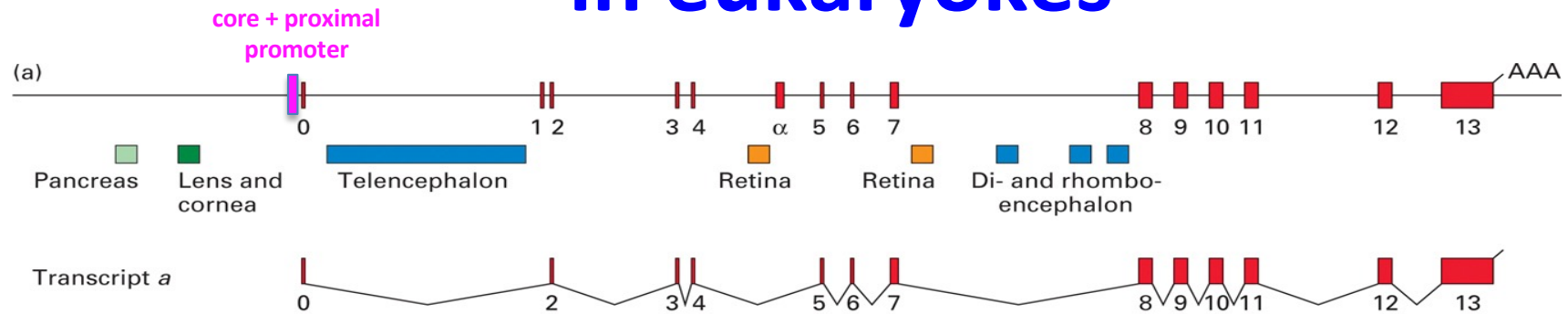
Efficient transcriptional activation often requires the combined and coordinated activities of several transcription factors bound to different enhancers to recruit RNAPol II

Though they may be distant in the linear DNA sequence, the DNA is bent and looped such that transcription factor-bound enhancers are physically located near the promoter

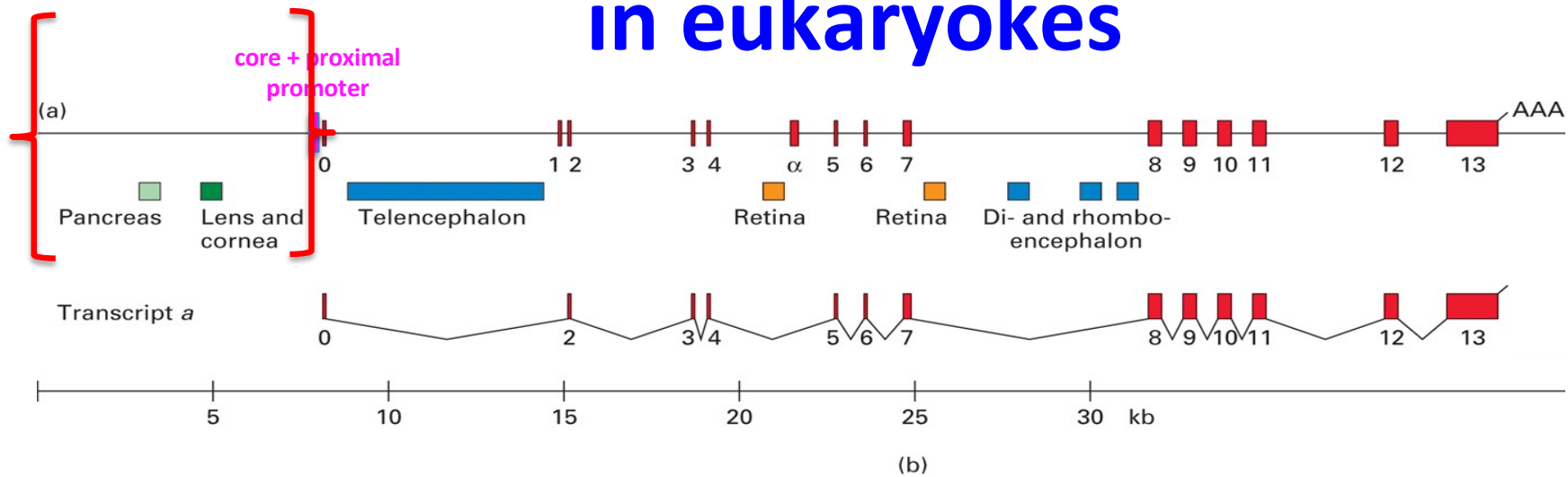
Testing for presence of enhancers in DNA: Reporter gene assays



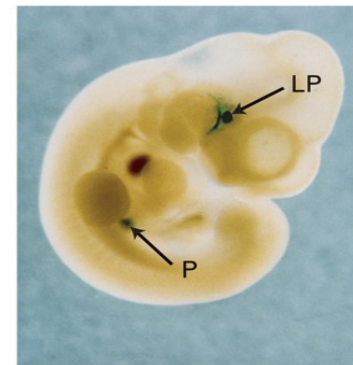
Complex regulation of transcription in eukaryotes



Complex regulation of transcription in eukaryotes



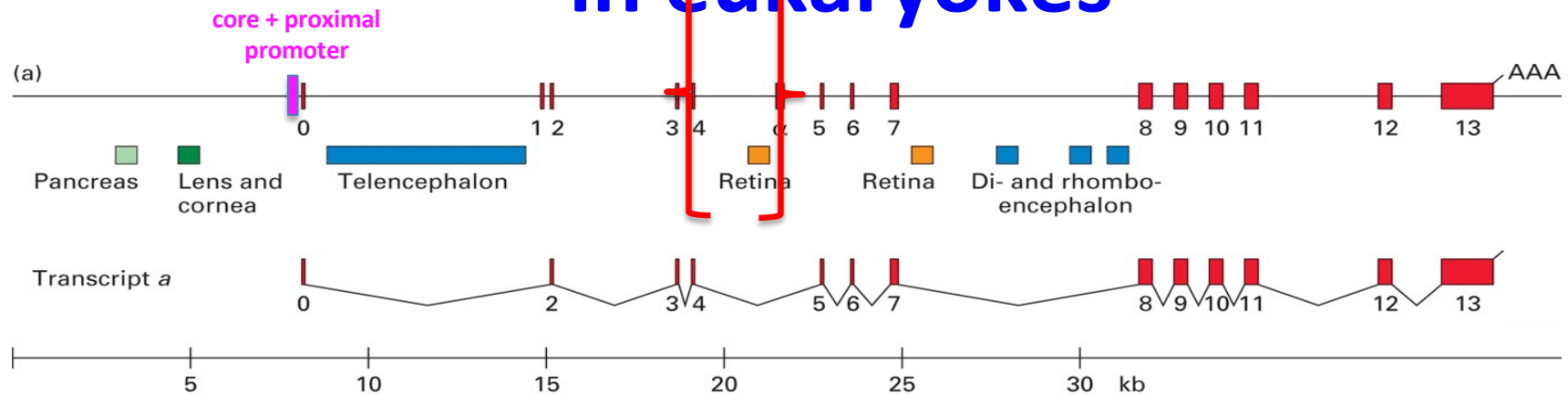
Pax6 enhancer
reporter gene assays :



From 0 to -8kb

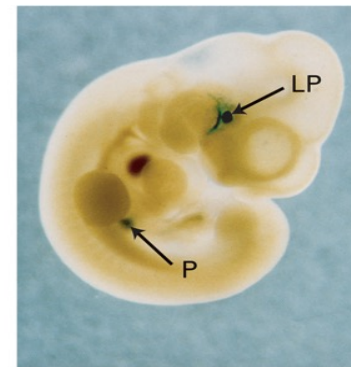
Lodish Ch7

Complex regulation of transcription in eukaryotes



Pax6 enhancer
reporter gene assays :

(b)



From 0 to -8kb

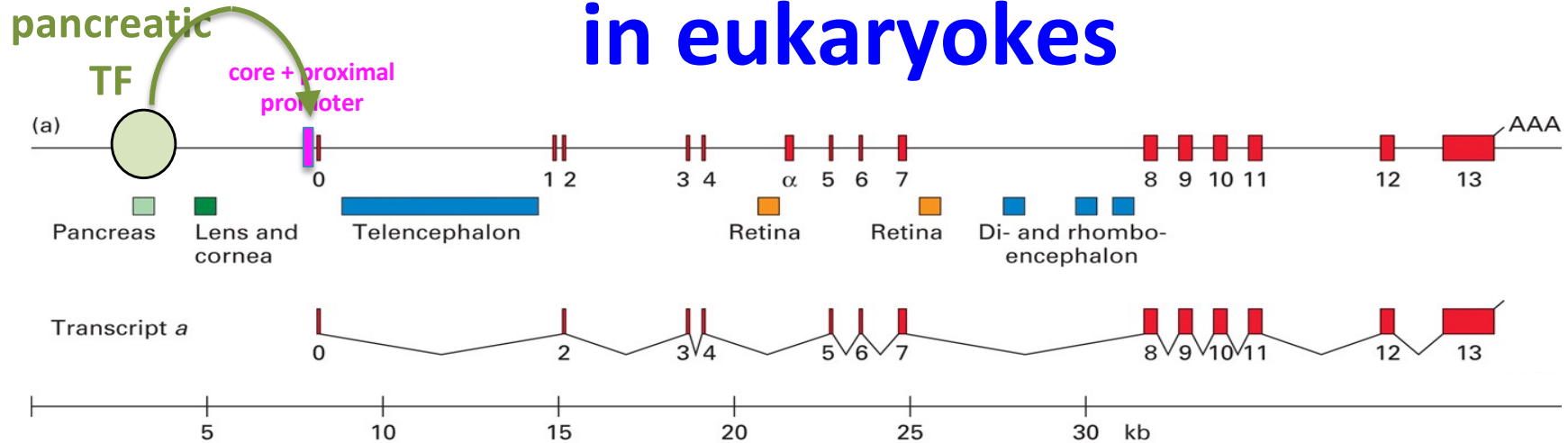
(c)



From Exon 4 to 5

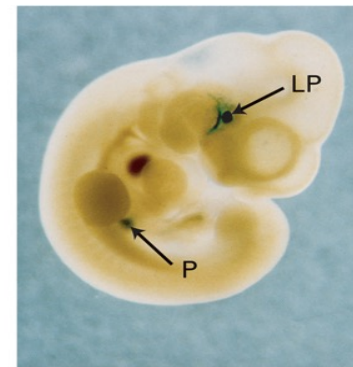
Lodish Ch7

Complex regulation of transcription in eukaryotes



Pax6 enhancer reporter gene assays :

(b)



From 0 to -8kb

(c)

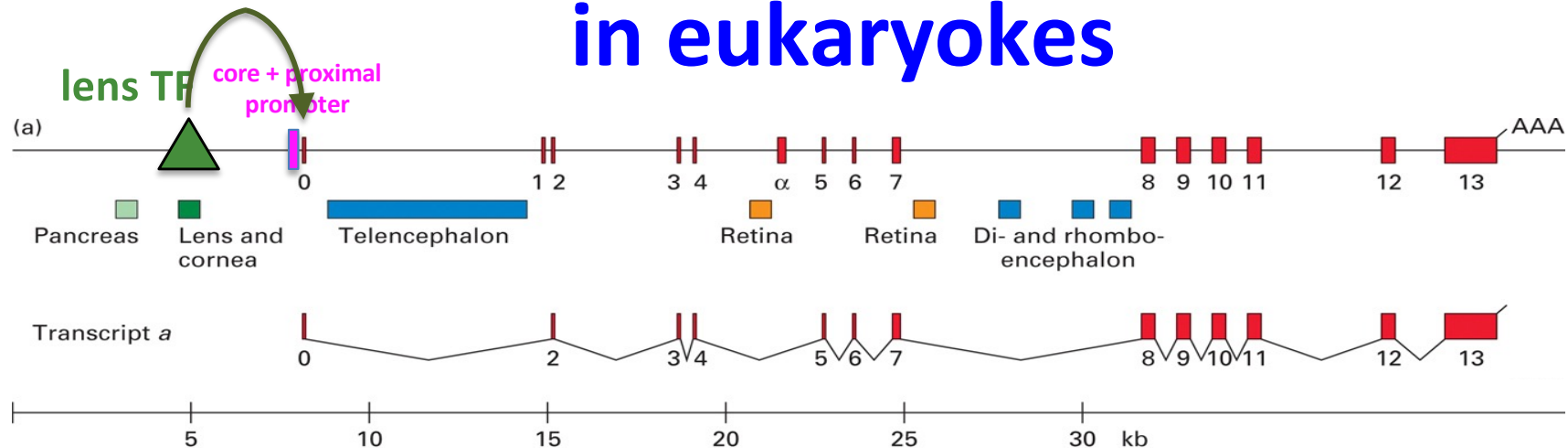


From Exon 4 to 5

Lodish Ch7

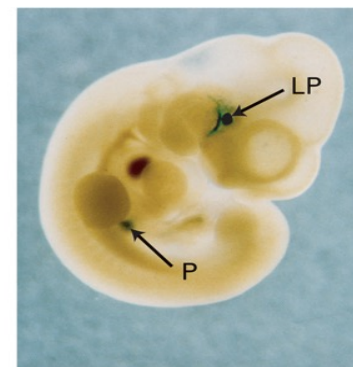
Different enhancers regulate PAX6 expression in specific tissues, an example of transcriptional complexity in eukaryotes

Complex regulation of transcription in eukaryotes



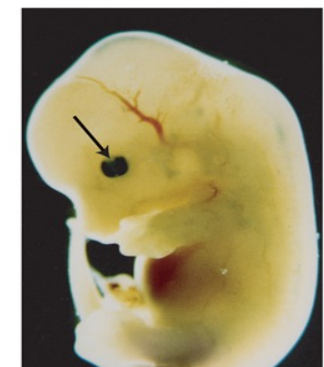
Pax6 enhancer
reporter gene assays :

(b)



From 0 to -8kb

(c)

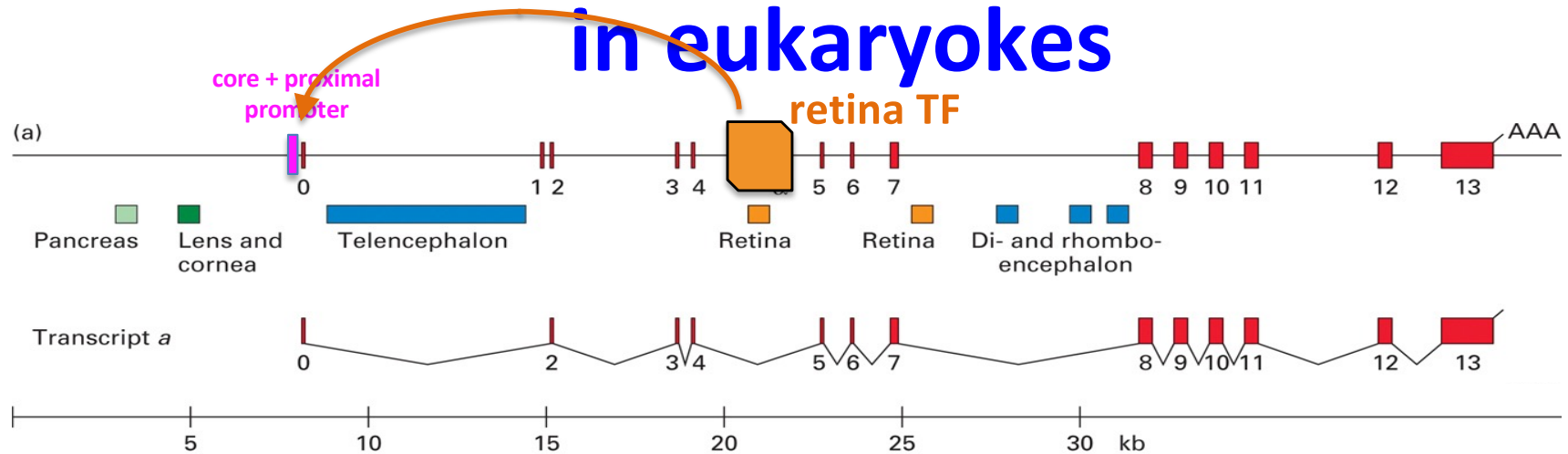


From Exon 4 to 5

Lodish Ch7

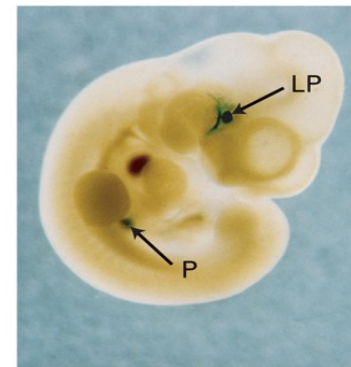
Different enhancers regulate PAX6 expression in specific tissues, an example of transcriptional complexity in eukaryotes

Complex regulation of transcription in eukaryotes



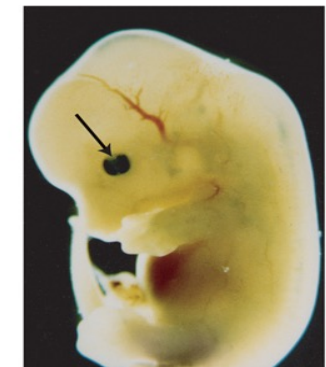
Pax6 enhancer
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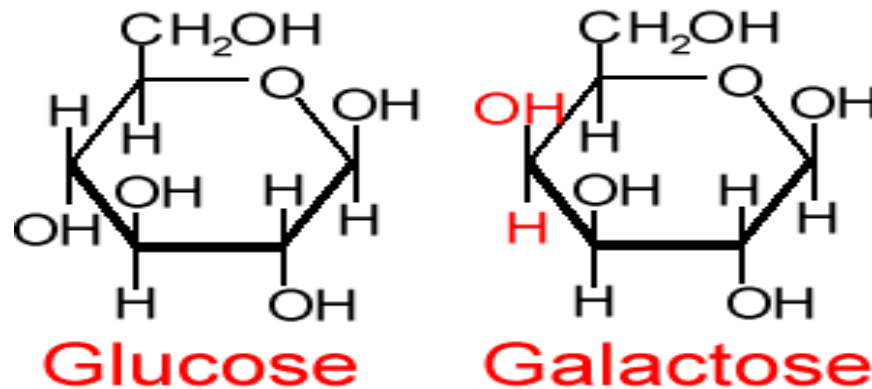
Different enhancers regulate PAX6 expression in specific tissues, an example of transcriptional complexity in eukaryotes

Researchers detect *Drosophila dpp* mRNA in cells in the epidermis and in the developing gut during embryonic stages. To see how this is regulated, they placed a 1 kb DNA fragment from intron 2 of the *dpp* gene upstream of GFP coding sequences and made embryos carrying this reporter gene in all their cells. They observed GFP fluorescence in the embryonic gut and nowhere else in these embryos. Which of the following statements is most likely TRUE?

- A. There are two different copies of the *dpp* gene in the *Drosophila* genome, each expressed in a different tissue
- B. Epidermis and gut cells must express the same transcription factor that activates *dpp* expression
- C. A transcriptional activating complex (enhanceosome) forms on intron 2 of the *dpp* gene in embryonic gut cells and not epidermis cells
- D. There is just one enhancer controlling activation of the *dpp* gene, located in intron 2

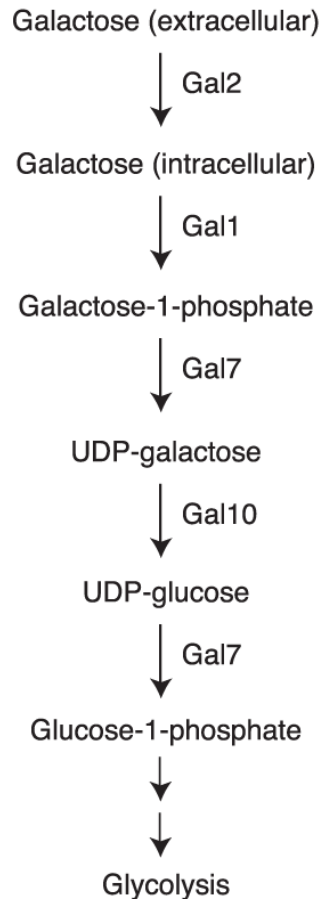
The yeast GAL system

- Molecular formula of glucose and galactose are identical. The difference between them is the orientation of the hydroxyl group at the 4th Carbon.
- In most living cells, glucose is the preferred monosaccharide sugar used in metabolism.
- Many species including yeast, which is a single cell eukaryote, can convert galactose to glucose-1-P for energy and carbon metabolism.



The yeast GAL system

The Gal pathway



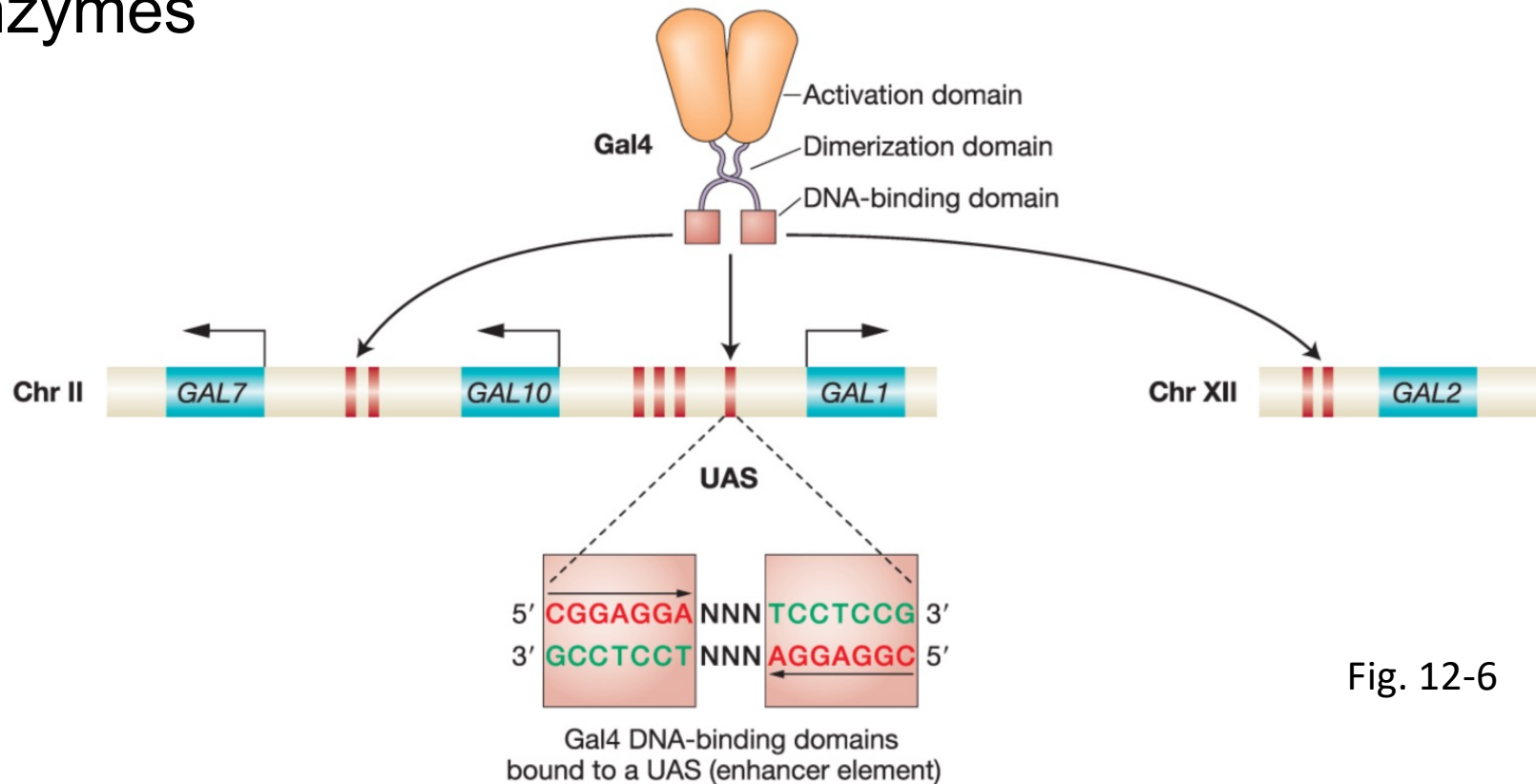
- Like the *lac* genes of *E. coli*, enzymes that convert galactose to glucose-1-P are only produced when galactose is present.
- Four galactose-responsive enzymes are required for this process (GAL1, -2, -7 and -10).
- Three regulatory proteins (GAL3, -4 and -80) are also required.

Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020
W. H. Freeman and Company

Fig. 12-5

GAL4 is a transcriptional activator

- GAL4 binds to enhancers called “Upstream Activator Sequences” (UAS) located upstream of each of the GAL enzymes



- each Gal gene has its own promoter - not organized into an operon
- UAS sites are located at a distance from the promoters, not immediately adjacent.

Control of GAL4 activity by galactose

Mutational analysis:

Gal4⁻ cells – Gal enzymes uninducible

Gal80⁻ cells – Gal enzymes constitutive

Gal3⁻ cells – Gal enzymes uninducible

- Gal80 binds to Gal4 activation domain – blocks activation
- binding of galactose and ATP changes structure of Gal3, causing Gal3 to bind to Gal80 and remove it from Gal4 (i.e. Gal3 acts as a sensor and inducer).

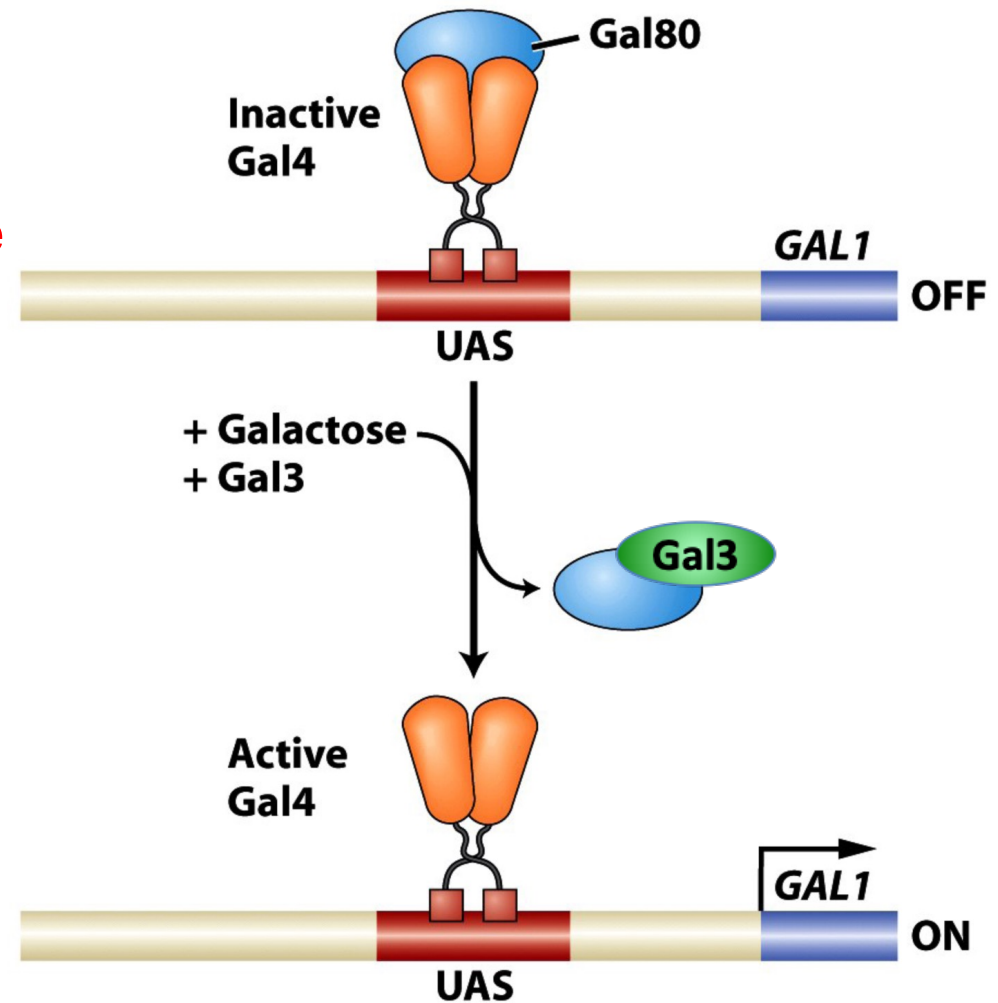
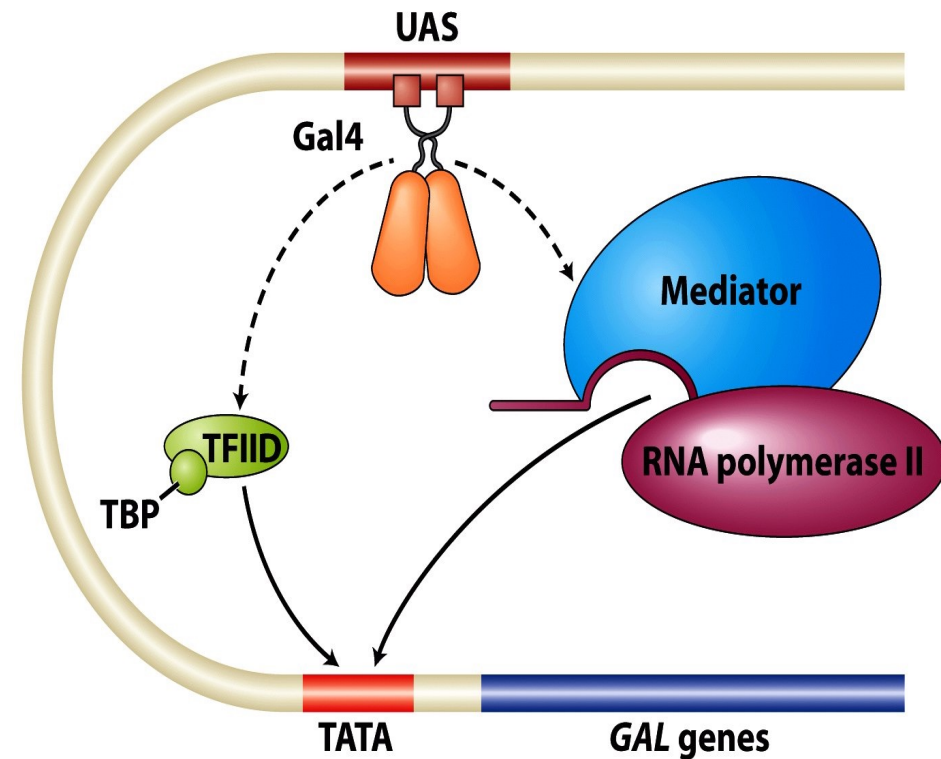


Fig. 12-8

Transcriptional activation by GAL4

The activation domain of Gal4 can bind to proteins of the transcriptional machinery:

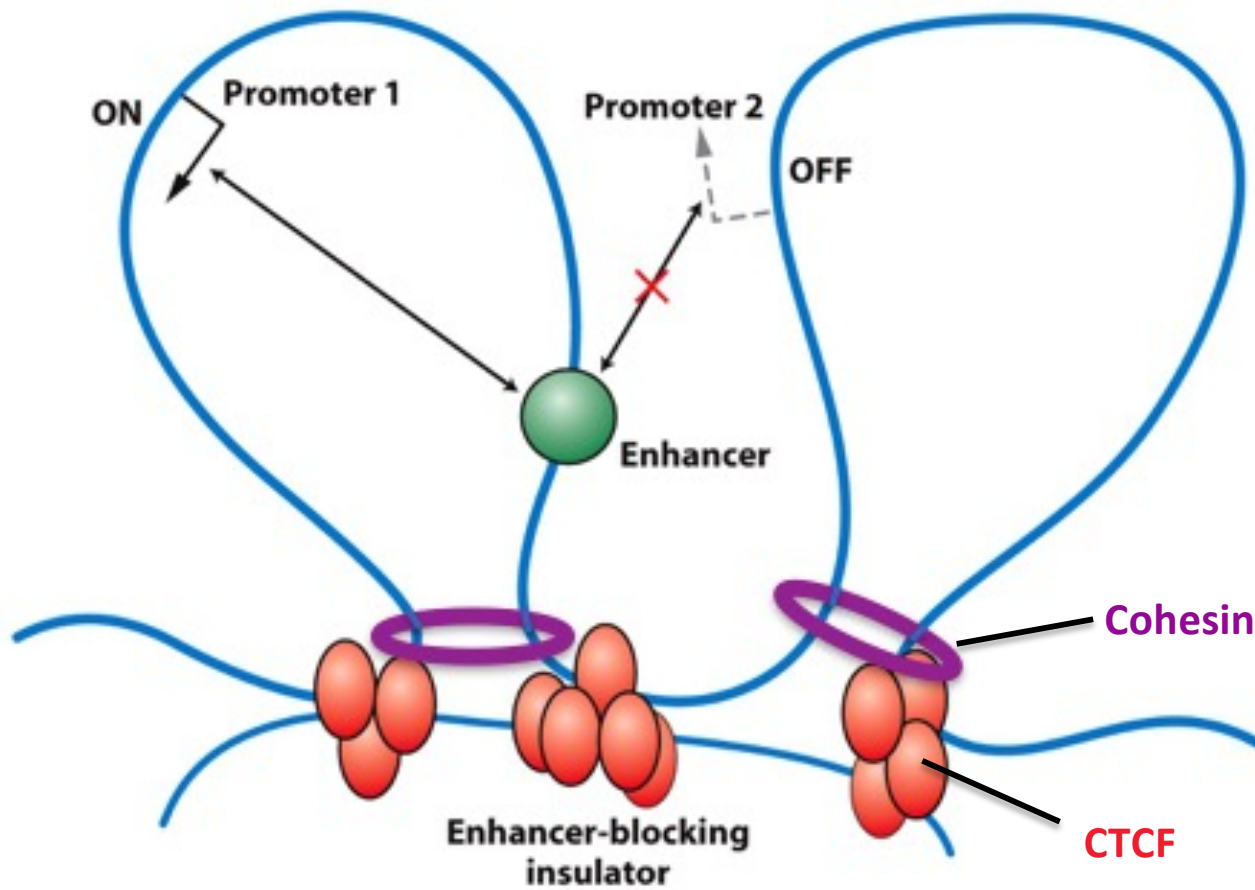
- helps recruit RNA Pol II
- also helps recruit chromatin modifying proteins
- GAL4 activation domain can function in many eukaryotic cells, insect, human, etc.



11th Edition Fig. 12-9

Enhancer-blocking insulators restrict the action of enhancers to specific genes

Topologically associating domain
(TAD)



Insulators:

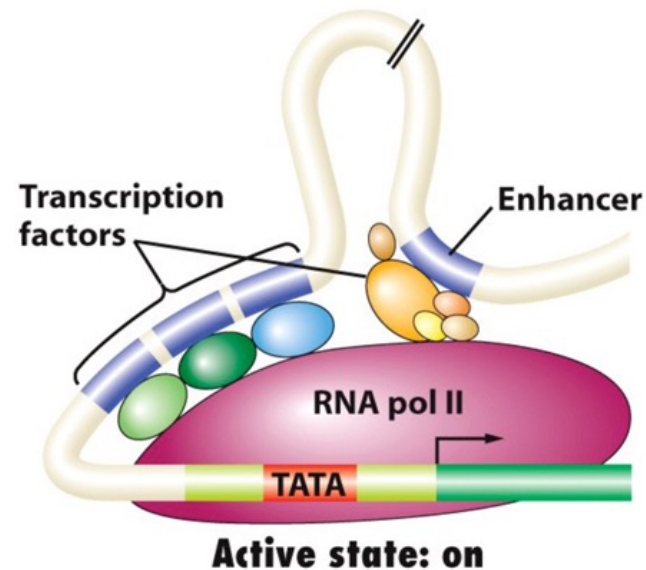
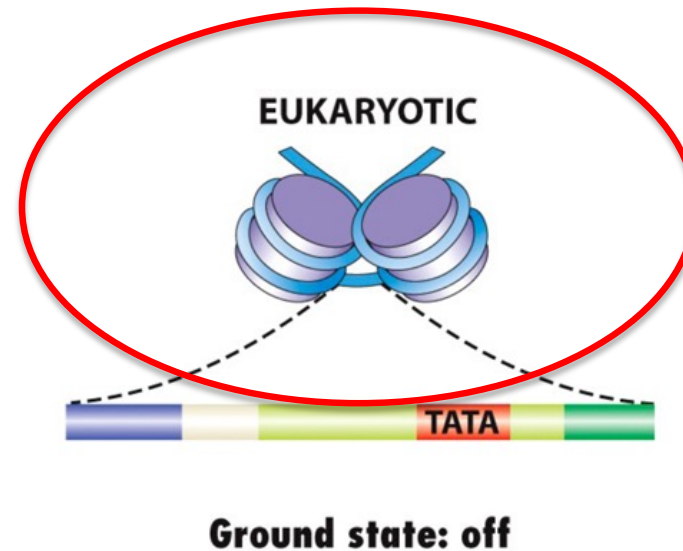
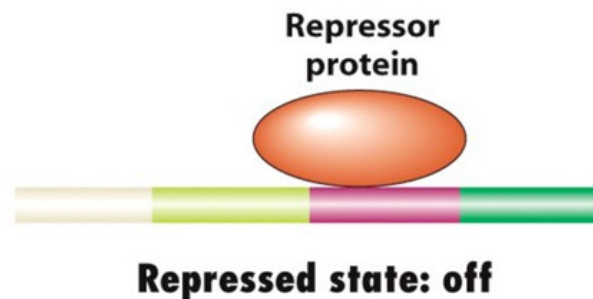
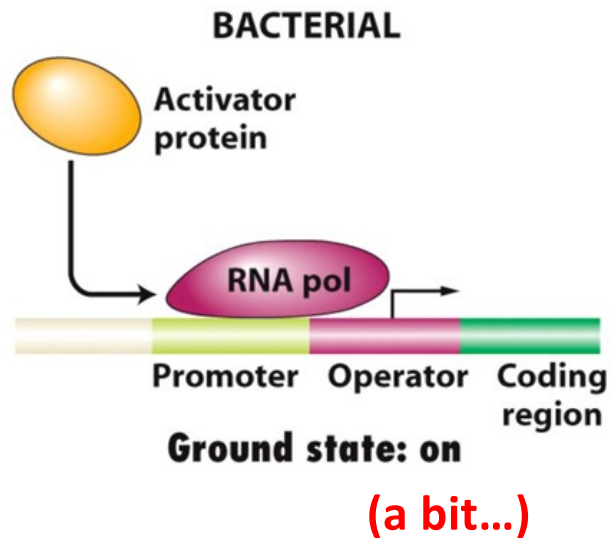
- DNA sequences that are specifically bound by "insulator proteins" like CTCF
- help to organize chromosomal DNA into specific sub-domains (TADs) essential for proper gene regulation

see Fig. 12-12

Which of the following is a common regulatory feature of both the Lac operon in *E. coli* and the Gal system in yeast?

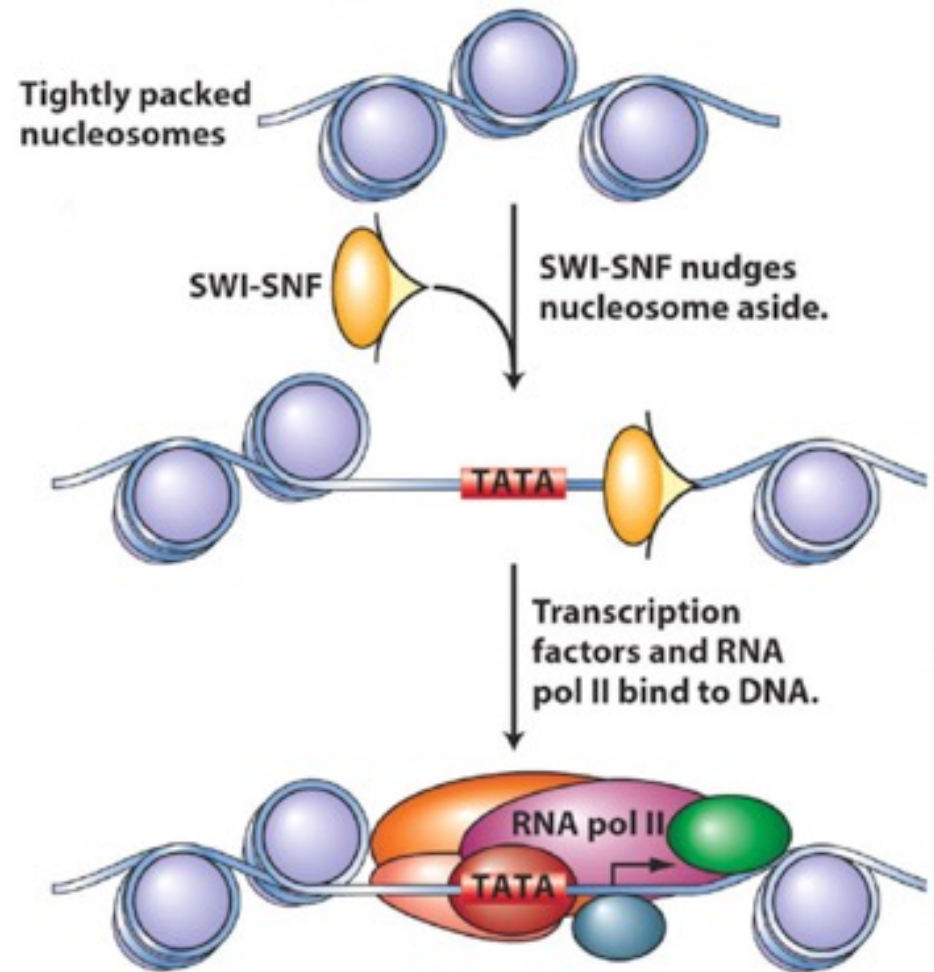
- A. the repressor binds to specific regulatory DNA sequences in the DNA
- B. binding of the sugar to its target protein causes an allosteric transition in the protein that alters its ability to bind DNA
- C. The transcriptional activator binds near the promoter of the regulated gene(s)
- D. Sequence-specific binding of the transcriptional activator to DNA is not sufficient to induce high-level expression of target genes

But what about the chromatin?



Chromatin often needs to be remodelled locally to allow gene transcription

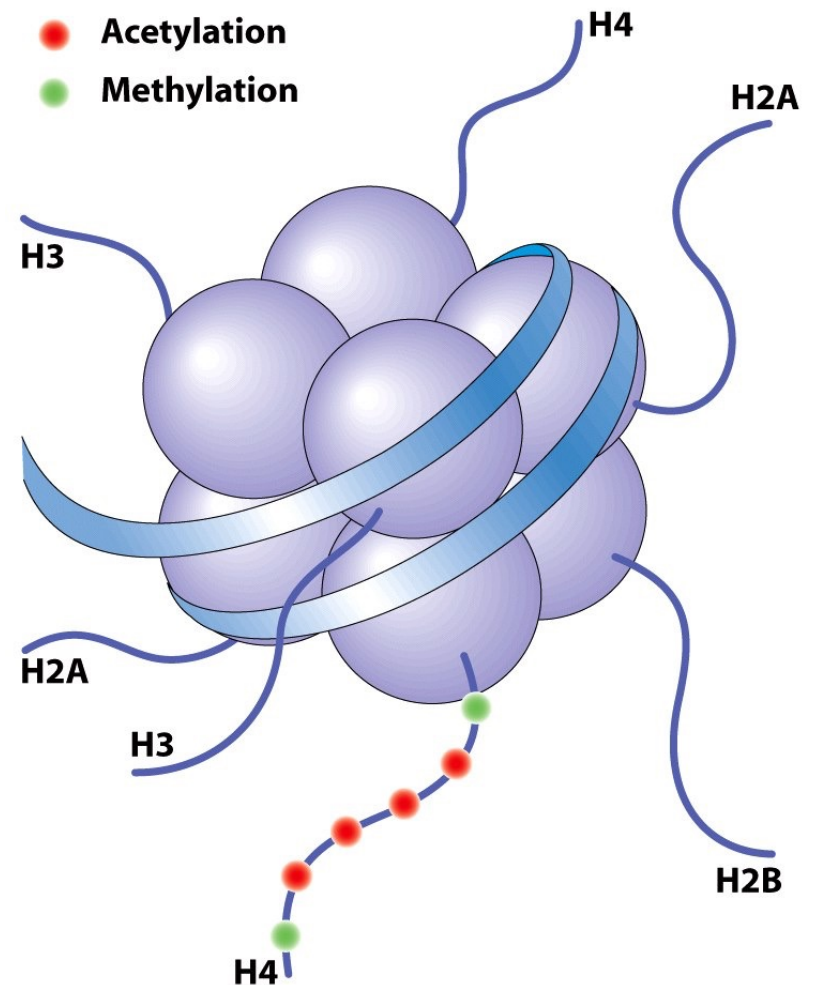
- transcription factor binding sites are often inaccessible due to their incorporation in nucleosomes
- nucleosome remodelling protein complexes like Swi/Snf use energy from ATP to reposition or remove single nucleosomes, exposing binding sites
- Swi/Snf is a **coactivator** - does not bind DNA itself, but gets recruited by enhancer-bound transcription factors



Histone modifications

- the amino-terminal ends of core histones protrude from the nucleosome
- variety of covalent modifications of these tails can have profound effects on gene regulation:

- **acetylation**
 - **methylation**
 - phosphorylation
 - ubiquitination
- can alter charge to affect affinity for DNA
 - can also affect the binding of other regulatory proteins (histone code)

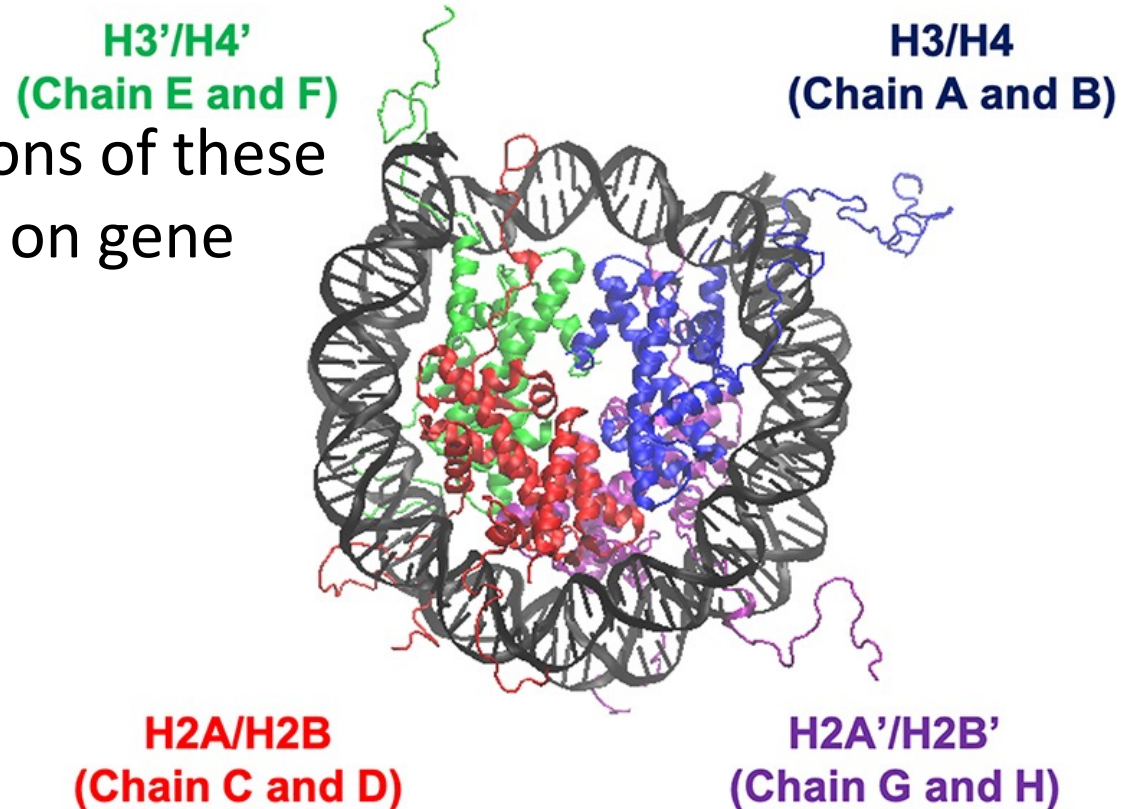


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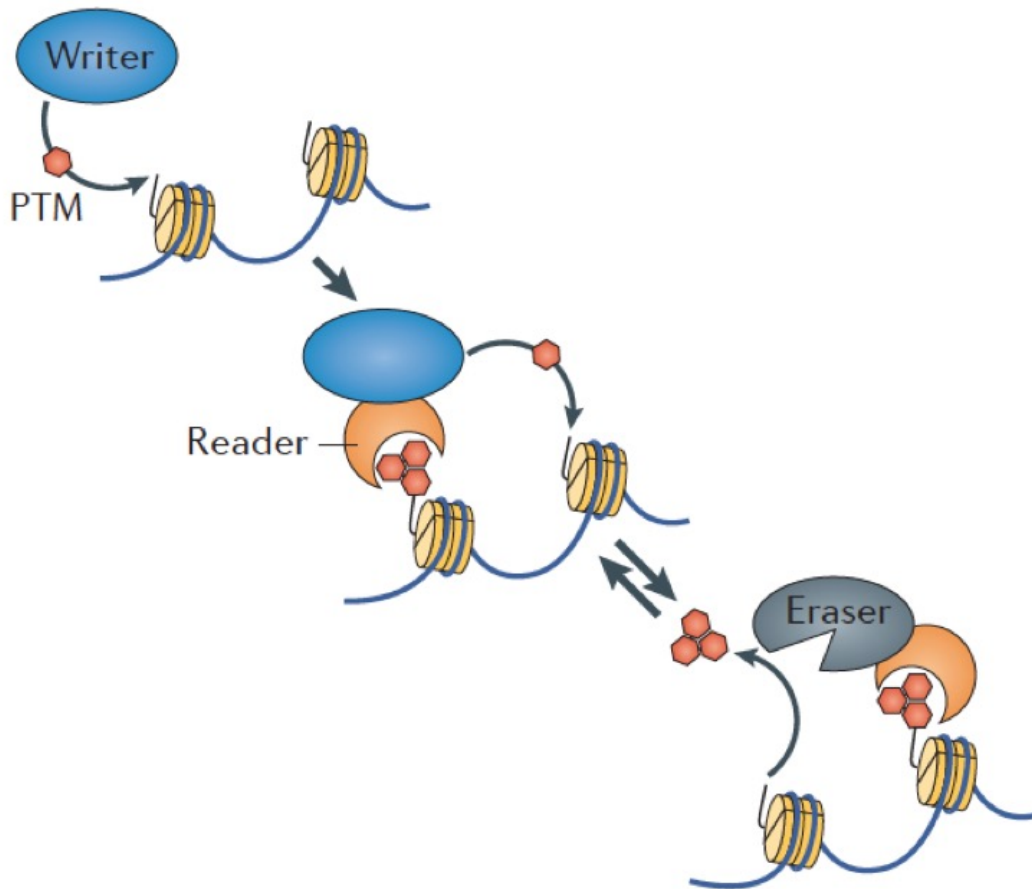
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See Fig. 12-10

Regulation of gene expression by histone modification: “the histone code”

c Reader–writer and reader–eraser coupling

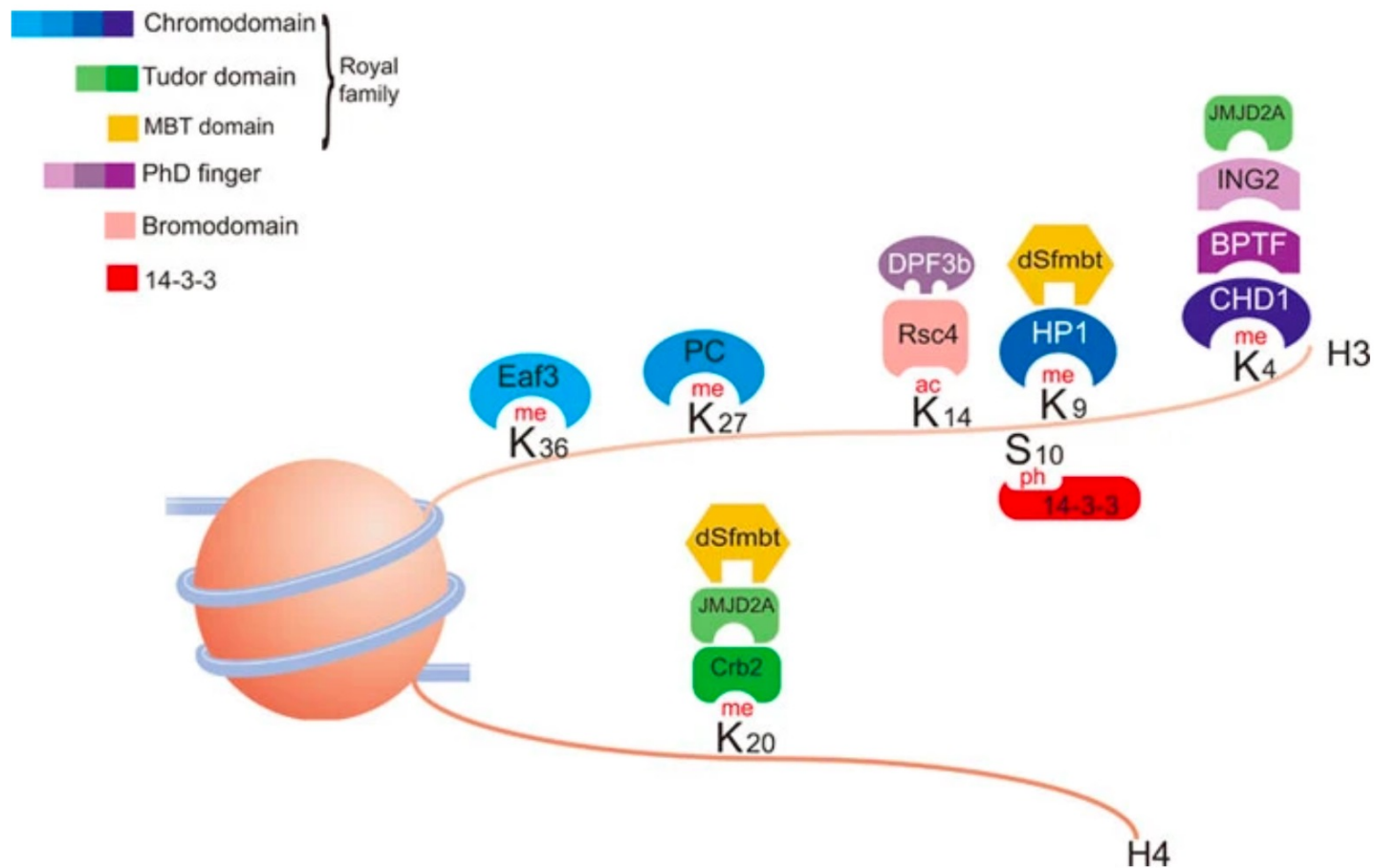


- Histone modifications are added by histone code **WRITERS**
- Histone modifications are removed by histone code **ERASERS**
- Histone modifications are recognized and bound by histone code **READERS**
- **readers are often coupled with writers or erasers (same protein or by interaction of two proteins)**

Allshire & Madhani (2017). *Nature Rev Mol Cell Biol*

See also Fig. 12-14

Different “readers” recognize different elements of the histone code

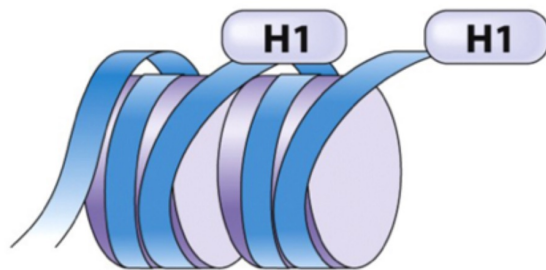


Bannister & Kouzarides (2011) Cell Research

See also Fig. 12-16

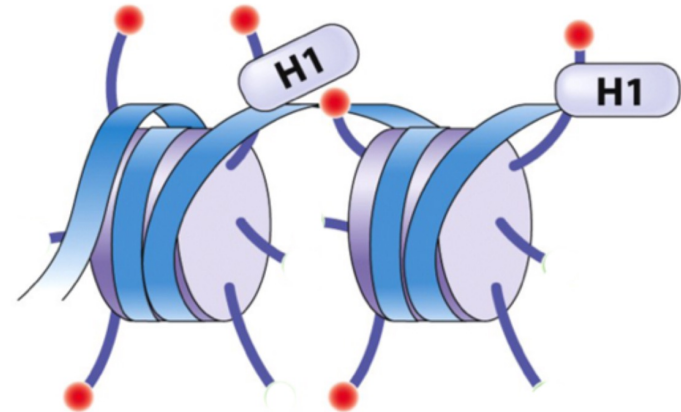
Histone acetylation relaxes chromatin and is associated with increased transcription

(a) Condensed chromatin



● Acetylation

(b) Open chromatin



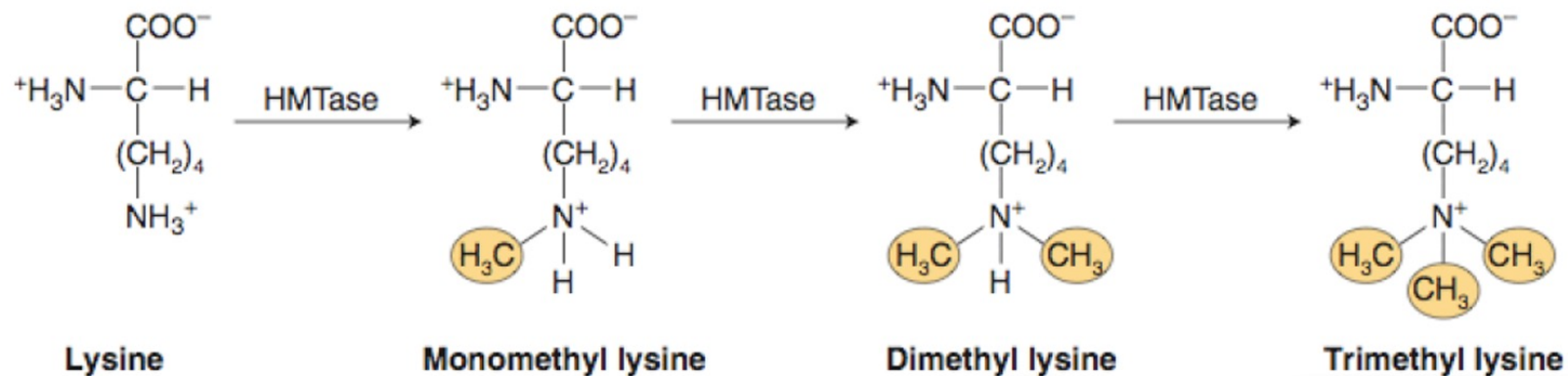
HAT
→
←
HDAC

11th Edition Fig. 12-14

- Histone tail acetylation on lysines neutralizes positive charge, loosens interactions with DNA (negative charge) → chromatin relaxation → increased accessibility and transcription
- Also creates binding sites for histone code readers that help promote activation
- Acetylation catalyzed by **histone acetyltransferase (HAT)** enzymes (“writers”); transcriptional (co)activators often have this activity
- Removed by **histone deacetylase (HDAC)** enzymes (“erasers”); transcriptional (co)repressors often have this activity

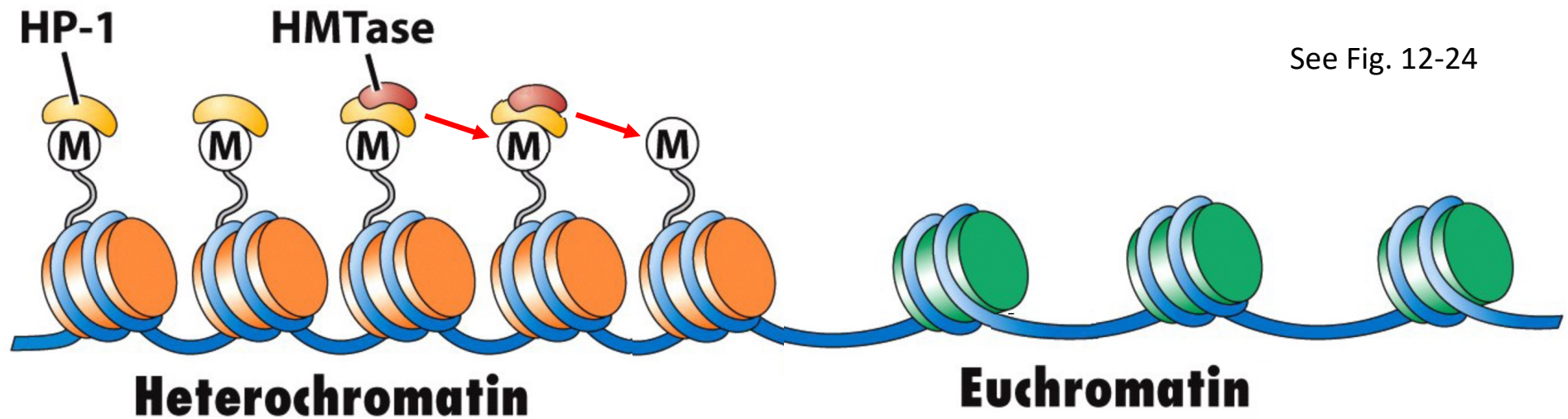
Histone tails can also be methylated

- Histone methylation occurs at lysine or arginine residues, catalyzed by **histone methyltransferase (HMTase)** enzymes
- Doesn't change the charge of histone



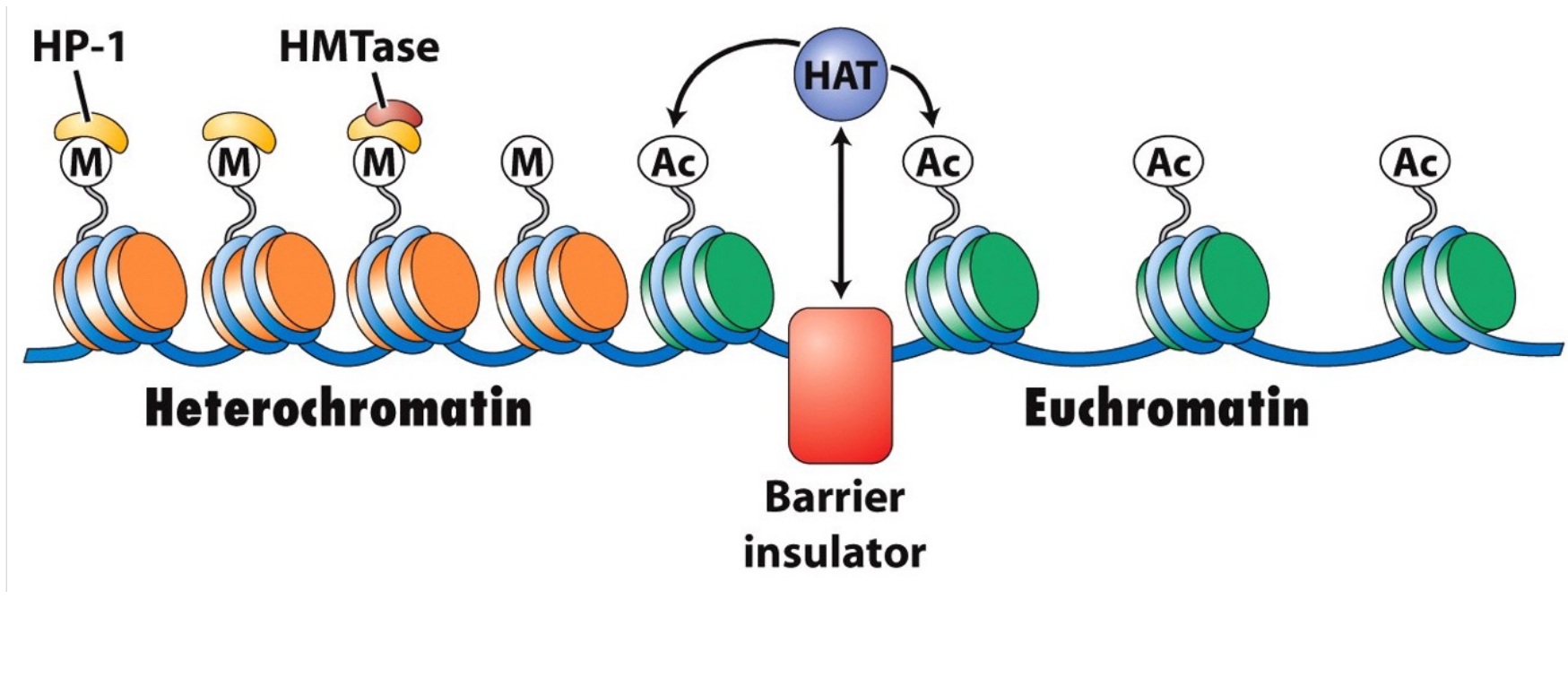
- Many histone methylation marks (e.g. H3K9Me) are generally associated with gene silencing and act as a signal to recruit specific readers.

Methylation (H3K9Me) promotes heterochromatin formation and tends to spread on chromatin



- HP-1 (Heterochromatin Protein 1) binds methylated histones, H3K9Me
- HP-1 promotes heterochromatin formation and recruits additional HMTase
- HMTase methylates neighbouring nucleosomes
- etc. etc.

Barrier insulators stop the spread of heterochromatin



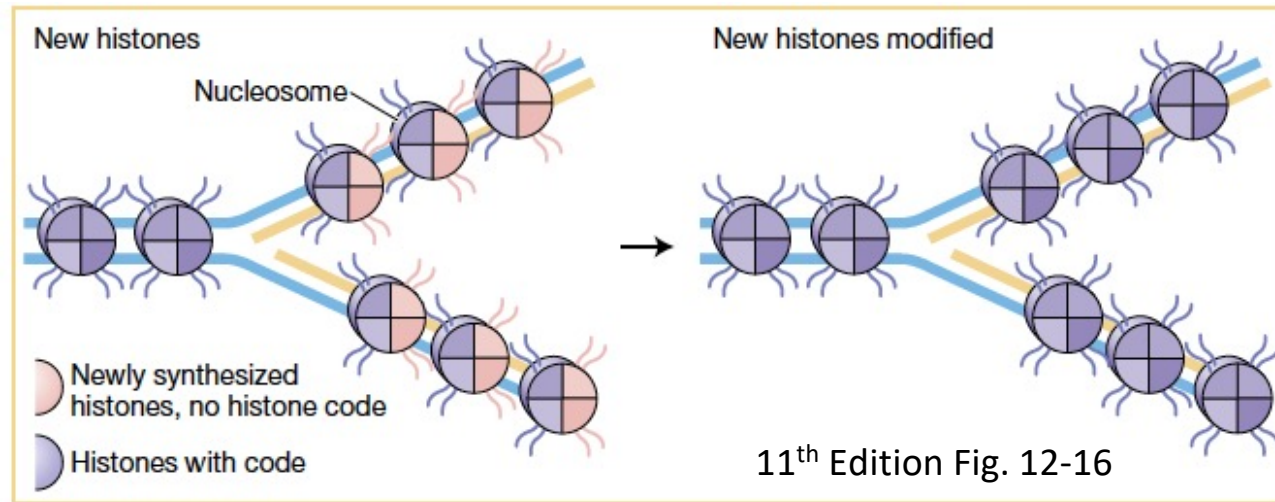
- heterochromatin spreading is counteracted by HATs bound to **barrier insulators (= DNA sequence)**

Which of the following changes is most likely to be associated with increased expression of the *α -tubulin* gene?

- A. de-acetylation of histones around the *α -tubulin* gene
- B. accumulation of heterochromatin protein-1 (HP-1) at the *α -tubulin* gene
- C. mutation in a gene encoding a histone methyltransferase that regulates *α -tubulin* expression
- D. mutation of a barrier insulator sequence located near the gene
- E. mutation in a gene encoding a histone acetyltransferase that regulates *α -tubulin* expression

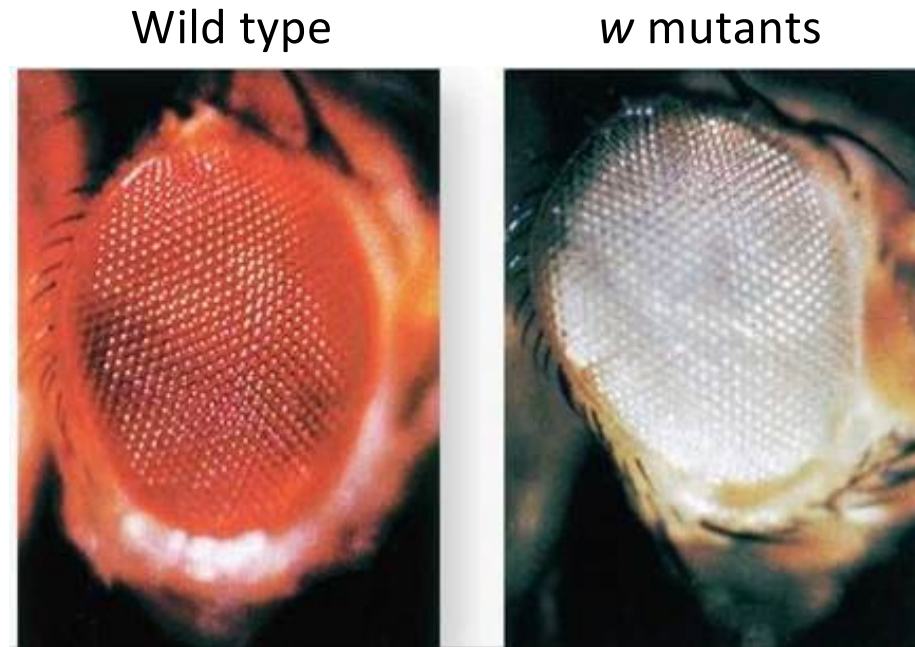
Epigenetic inheritance of chromosome states

- **Epigenetics**: study of heritable traits that cannot be explained by changes in DNA sequence.
- Many modifications to DNA and chromatin structure are inherited: passed on to daughter cells or future generations.



- During DNA replication, existing nucleosomes are disassembled and new nucleosomes on daughter strands are reassembled on the daughter strands from a mix of old and new (unmodified) histones
- Old histones with modifications (methylation) direct modification of new histones by recruiting readers and writers

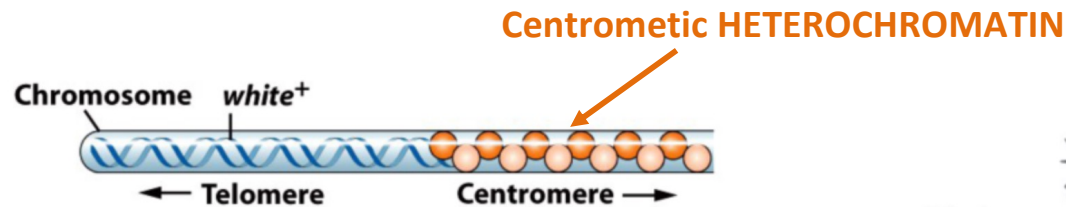
An epigenetic phenomenon in the *Drosophila* eye



Two classes of pigments, brown and red pigments, contribute to the *Drosophila* eye color.

The “white” (*w*) gene codes an ATP-binding cassette (ABC) transporter that carries the precursors of the red and brown eye color pigments into the developing eyes

Position-effect variegation: Epigenetic silencing by heterochromatin spreading



Mutagenize flies with X-rays
Screen for offspring with unusual phenotype

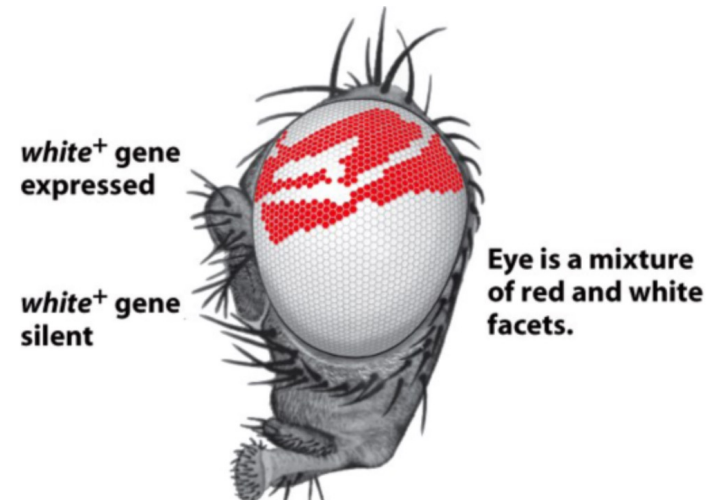
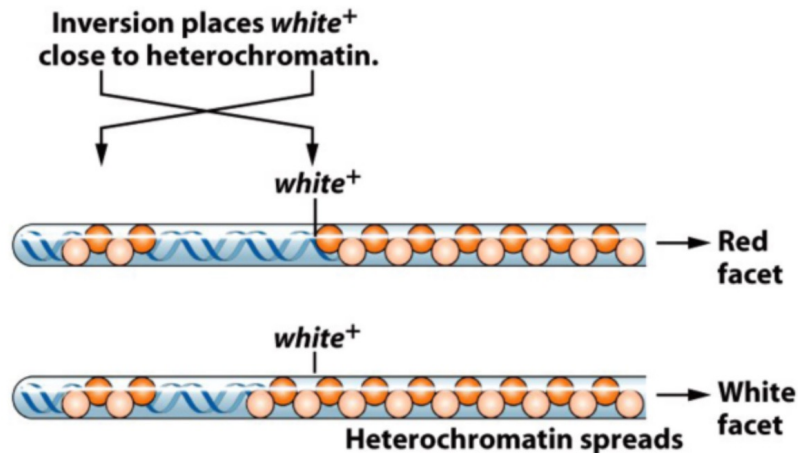
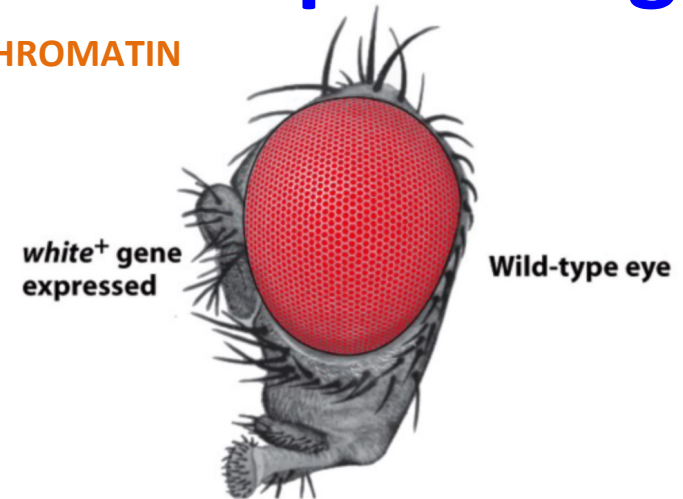


Fig. 12-22

Identification of proteins that promote or prevent the spread of heterochromatin

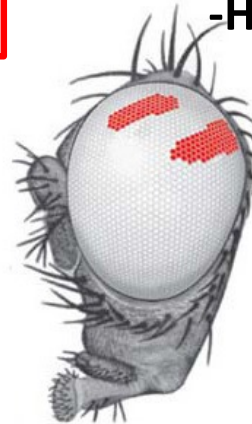
Second-site mutations that affect the spreading of heterochromatin

Examples of mutation identified (LOF):

-HAT: Histone acetyltransferase

Drosophila eye
(translocated *white*⁺)

E(var)



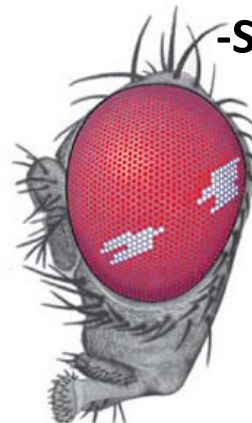
Spreading enhanced.
More *white*⁺ are silenced.

Examples of mutation identified (LOF):

-Su(var)3-9: Histone methyltransferase

-Su(var)205: HP1

Su(var)



Spreading suppressed.
Fewer *white*⁺ are silenced.

Fig. 12-23